cent. These results will afford independent temperature and density measurements for the active emission region and will also yield abundance data for those elements the lines of which are measured. In particular, the neon line measurements are likely to afford an improved estimate of the coronal neon abundance. Further, a comparison of the neon X 12·13 Å flux with the neon IX 13·45 Å intensity will provide a separate and sensitive temperature determination. In this connexion it is particularly important to ascertain what contribution the iron XVII 12·12 Å transition makes to the emission observed at 12·13 Å.

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PLANETARY SCIENCE

Devonian Palaeomagnetic Pole for Europe and N. America

The position of the Devonian pole relative to Britain, was originally deduced from palaeomagnetic measurements on the Old Red Sandstone formation from S. Wales[1]. It was apparently confirmed later by measurements on many other Devonian formations from Europe and western Russia[2]. We label these poles "DI" in Table 1. Palaeomagnetic work on Devonian lavas from the Midland Valley of Scotland yields, however, a completely different pole[3] which is labelled "DII" in Table 1 (ref. 4).

Pole DI was accepted as the European Devonian pole without question for many years, largely because of the large number of formations which yield it. These formations are almost exclusively sedimentary, however, and their natural remanence is quite possibly younger than the rocks themselves or their fossil content. This possibility prompted Chamalaun and Creer[5–7] to carry out thermal demagnetization experiments on the Old Red Sandstone. In most of the samples examined, a small fraction of the natural remanence was found to have blockage temperatures higher than 600° C and yielded pole DII. Pole DI is close to the European Permo-Carboniferous pole (see Table 1), and so it was argued that the Old Red Sandstone must have acquired most of its remanent magnetization in the Permo-Carboniferous, and this could have been caused by physical and chemical changes in the iron oxide content. Pole DII was thus tentatively accepted as the true Devonian pole for Britain.

Another argument in favour of the remagnetization hypothesis is that when pole DII is accepted as the Devonian pole for Europe, an upper Palaeozoic world geography similar to Wegener's[8] can be reconstructed using world-wide palaeomagnetic data[9,10]. No sensible continental reconstruction can be made, however, if pole DI is accepted as the Devonian pole for Europe.

There is also evidence of two different groups of poles for North American Palaeozoic rocks[9,10], one set corresponding to the European DI and the other to DII. Poles from the DII group had to be used to develop the palaeomagnetic continental reconstruction. It follows that the Permo-Carboniferous remagnetization must have been widespread and that it must have affected Devonian (and also younger Palaeozoic rocks) all over Europe, West Russia and North America.

Satisfactory thermal demagnetization experiments have not yet been carried out on many of these geographically widely distributed Devonian red beds magnetized in the DI direction. In experiments on two Norwegian formations[11], pole DII was not found, although the cleaned direction yielded a pole displaced from DI towards DII. This could be caused by almost complete remagnetization: this single negative result does not invalidate the remagnetization hypothesis. It is thus appropriate to tackle the problem from the other direction, that is, to examine again the palaeomagnetism of the Scottish lavas, because Stubbs's measurements were carried out before cleaning

Table 1.

No.	Formation	Treatment	No. of sites	Palaeomagnetic N. Pole Latitude	Longitude		Group
1	Old Red Sandstone sediments (Anglo–Welsh Cuvette)	n.r.m.	35 sites	39° N.	155° E.	4°	DI
2	Lower Devonian lavas (Midland Valley, Scotland)	selected n.r.m.	8 sites } 8 sites }	6° S.	142° E.	9°	DII
3*a*	Old Red Sandstone sediments (Cwmbran, South Wales)	n.r.m.	49 samples	40° N.	173° E.	6°	DI
3*b*	Old Red Sandstone sediments (Cwmbran, South Wales)	n.r.m.	1 site (5 samples)	1° S.	119° E.	6°	DII
4	Old Red Sandstone sediments (Anglo–Welsh Cuvette)	thermally demagnetized	5 sites	2° S.	117° E.	19°	DII
5	Lower Devonian lavas (Midland Valley and Lorne Plateau, Scotland)	a.c. cleaned	8 sites } 2 sites }	1° N.	121° E.	11°	DII
6	Permian			43° N.	165° E.	4°	
7	Carboniferous			35° N.	146° E.	9°	

n.r.m., Natural remanent magnetization.

Table 2. SUMMARY OF PALAEOMAGNETIC DATA FROM SCOTTISH FLOWS

Flow	Location	N	Directions of remanent magnetization D	I		Palaeomagnetic N. Pole Latitude	Longitude	Sign of field
L4	Tayside, Fifeshire	7	52·1	− 6·3	−	17·0° N.	119·5° E.	N
L8	,, ,,	8	259·2	+49·1	14·2	19·1° S.	110·8° E.	R
L10	Stonehaven–Montrose area	8	212·0	+27·5	16·5	14·1° N.	143·1° E.	R
L13	,, ,, ,,	12	39·1	−47·1	3·4	1·0° S.	141·3° E.	N
L14	,, ,, ,,	11	53·2	−47·7	5·0	6·4° S.	130·1° E.	N
L15	Lorne Plateau, Argyllshire	8	63·8	−11·1	10·3	9·3° N.	110·2° E.	N
L16	,, ,, ,,	8	70·4	−21·4	−	1·2° N.	107·4° E.	N
L18	,, ,, ,,	11	65·6	−25·7	8·5	1·5° N.	112·7° E.	N
L19	,, ,, ,,	12	68·2	−15·6	4·9	5·0° N.	107·6° E.	N
L20	,, ,, ,,	28	56·9	−53·4	1·3	12·5° S.	129·6° E.	N

All flows are Lower Devonian in age. Directions were calculated after treatment in a.c. fields up to 255 oersteds (peak value).

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techniques had been developed. He did, however, adopt strict criteria of acceptance of data[3].

In this context we now report results of a new palaeomagnetic survey of the Scottish lavas on which a.c. and thermal cleaning have been carried out[12]. These results are summarized in Table 2, and will be published in full elsewhere. The new mean direction of magnetization and its pole are listed as entry five in Table 1, and plotted as point five in Figs. 1*a* and 1*b*.

There is very good agreement between the cleaned DII poles. That for the cleaned lavas (No. 5) is very close to that for the thermally cleaned Old Red Sandstone from South Wales (pole 4). Both these poles agree well with No. 3*b* derived from a minor number of Old Red Sandstone sediments from Cwmbran, South Wales, in which we suggest the primary Devonian direction DII has remained predominant. The majority of beds from this site are thus supposed to have been remagnetized in direction DI (entry 3*a*).

The results of this new palaeomagnetic survey of the Devonian lavas strengthen the argument that DII is the true Devonian pole relative to the United Kingdom and validates the criteria of stability used by Stubbs in his original work.

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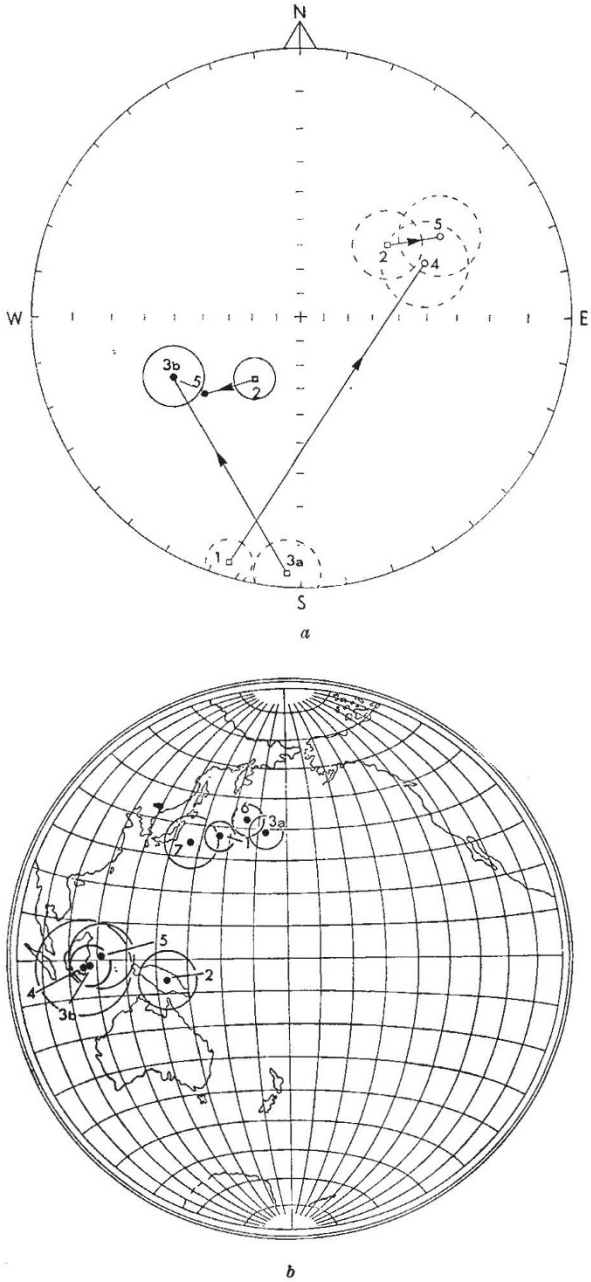


Fig. 1. *a*, Mean directions of magnetization of the Old Red Sandstone and Lower Devonian lavas. Point numbers refer to formations listed in Table 1. Open circles indicate upward inclinations, closed circles indicate downward inclinations. *b*, Pole positions (listed in Table 1) computed from mean directions of magnetization shown in *a*.

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Seasonal Behaviour of Atmospheric Pressure Oscillations

It is well known that the atmospheric pressure at any given point does not remain constant in time and is subject to periodic and non-periodic fluctuations. Statistical analyses have been used to establish them and many investigations have been published. This communication reports the results of a spectral density analysis performed for the barometric pressure data at the Langmuir Laboratory (33° 59′ N., 107° 11′ W., elevation 3,320 m above mean sea level), a research facility of the New Mexico Institute of Mining and Technology, and at the Institute campus at Socorro (34° 4′ N., 106° 54′ W., elevation 1,410 m above mean sea level). The Langmuir Laboratory is on a summit ridge of the Magdalena Mountains and the campus is 27 km east in the Riogrande Valley. A comparison of the results of these two stations provides an interesting example of atmospheric behaviour in a "mountain–valley" situation.

Barometric pressure records from both stations were sampled every 2 h throughout 1965. The year was divided into nine periods, eight of 40 days and the last one of 45 days. The analysis first estimates the autocovariance and then takes its Fourier transform to calculate the spectral density or power spectrum. The computations were carried out on an *IBM*–1620 computer, using a procedure described by Blackmann and Tukey[1]. Identical analyses were also carried out for the year 1964 and yielded similar results.

Figs. 1 and 2 illustrate respectively the spectral density estimates of barometric pressure at the mountain and Socorro. The spectra for both stations show that the power in general decreases with the increase in frequency. They indicate the presence of 24 h (diurnal), 12 h (semi-diurnal), 8 h (terdiurnal) and 6 h (quartodiurnal) period oscillations and show little evidence of oscillations of higher frequency than quartodiurnal.

There is a distinct seasonal effect on 24 h, 8 h, and 6 h waves, but none can be discerned on 12 h oscillations. The power associated with the semidiurnal wave is at a maximum in the third period (days 81–120) for the mountain and the second period (days 41–80) for Socorro. At both stations, the diurnal wave is at a minimum in the winter and at a maximum in the summer months. It virtually disappears at the mountain during winter. This

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